

A Framework for Rapid Prototyping of Virtual Reality Training Scenarios

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ABSTRACT

This paper introduces a robust framework designed to optimize the rapid prototyping of VR training applications, focusing on user experience and reusability. Developed from a case study on electrical risk training for a Hydroelectric Plant, this framework integrates essential systems for interaction, error detection, and objective tracking. Preliminary evaluations demonstrate its potential to significantly reduce development time and improve training outcomes.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality HCI design and evaluation methods—Heuristic evaluations

1 INTRODUCTION

Extended reality (XR) technologies, particularly virtual reality (VR), are increasingly applied in education, healthcare, and industrial training. With the advent of accessible devices like head-mounted displays (HMDs), industries are exploring VR's potential to simulate high-risk environments safely and cost-effectively. This paper explores a framework designed for rapid prototyping of VR training scenarios, emphasizing user experience and adaptability.

The work is motivated by a demand for a VR-based simulator prototype to study the feasibility of using VR in electrical risk for training purposes. Insights from this demand guided the framework's development, focusing on modular systems for interaction, error tracking, and adaptive user feedback [1].

2 RELATED WORK

Prior studies highlight the growing adoption of VR for training across various domains such as industry [8, 7, 9], medicine [6, 2], and emergencies [3]. VR for training reduces real-world risks by simulating hazardous scenarios, as demonstrated in industrial safety training [9]; in addition, the technology allows the integration of gaming elements to increase user engagement and retention, contributing to effective learning outcomes [4].

Some studies have focused on developing frameworks tailored to specific fields within VR training [5], with the purpose of reducing development time and providing reusable tools. Although these frameworks effectively address the needs of their respective domains, there is potential to design a more versatile solution that supports a wider range of training scenarios.

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3 FRAMEWORK DESIGN

The proposed framework tackles several common challenges associated with VR training development. First, it emphasizes intuitive and effective interaction systems by providing mechanisms for object manipulation and integrating seamless user interfaces. In parallel, the framework incorporates robust error detection and feedback procedures to track user actions, identify mistakes in real-time, and deliver corrective guidance. This real-time feedback loop helps users refine their performance and supports skill acquisition.

Furthermore, the framework ensures that training objectives are systematically met by dynamically adjusting tasks and offering structured assistance designed to optimize learning outcomes. From a development standpoint, the framework streamlines the creation of VR scenarios by modularizing essential systems, thereby reducing the time, cost, and resources required to bring effective training modules to fruition. Its design also prioritizes flexibility and adaptability, making it straightforward to customize for various industries and training contexts. Finally, the framework enhances the overall user experience by integrating user-friendly features and minimizing barriers to immersive learning, ultimately driving more impactful and efficient VR training initiatives.

4 DEVELOPMENT

Developed using Unreal Engine in version 4.27, with a combination of C++ and Blueprints, the framework incorporates modular components for easy customization. It supports integration with advanced VR peripherals like the HTC Vive Pro Eye and Meta Quest. Its key components are described as follows.

4.1 Main Character Representation

The representation of the main character within the VR environment was developed with a strong emphasis on realism and usability. Specifically, the system focuses on detailed hand representations that simulate natural movements. The physical constraints applied to the virtual hands ensure that movements are intuitive and align with the user's actions in the physical world. This realistic depiction enhances user immersion and facilitates accurate interactions.

4.2 Interaction System

The interaction system was designed to provide users with intuitive tools for object selection and manipulation within the virtual environment. This system leverages Hand Socket Components to enable precise alignment and gripping of objects. Collision detection mechanisms were refined to ensure accuracy during interactions, minimizing unintended actions. Additionally, hand animations dynamically adapt to the objects being manipulated, further enhancing the user's sense of presence and control. Figure 1 showcases the interaction flow, highlighting object *FingerTipSelection* processes.

4.3 Snap System

To simplify complex assembly tasks in the virtual space, a snap system was introduced. This system enables objects to align auto-

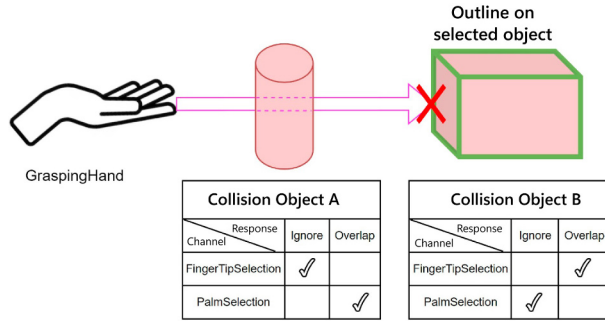


Figure 1: *FingerTipSelection* interaction flow example.

matically when brought into proximity with predefined snap points, ensuring correct placement. The snap system improves the efficiency of training scenarios requiring assembly or configuration tasks by reducing user errors and providing immediate visual feedback. Figure 2 illustrates how the implemented snap system works, based on a user-friendly design.

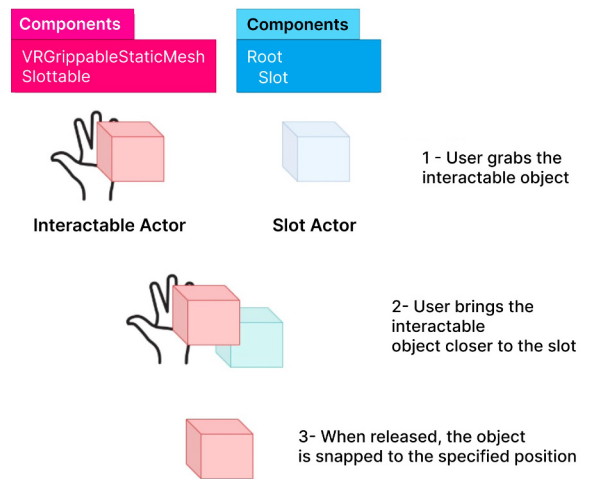


Figure 2: Snap process for an interactive object.

4.4 Locomotion System

A versatile locomotion system was implemented to accommodate diverse training scenarios and user preferences. This system includes teleportation for rapid movement and free movement for scenarios requiring continuous navigation. Visual indicators, such as dynamic markers, were added to guide users in selecting teleportation destinations, improving spatial awareness and navigation clarity. Figure 3 illustrates the visual indicators and their integration into the teleportation system.

4.5 Assistance System

The assistance system was developed to facilitate collaborative training experiences. An asymmetric multiplayer mode allows an instructor to provide real-time guidance to a trainee within the VR environment. The instructor interface includes tools for monitoring trainee actions, offering feedback, and dynamically adjusting

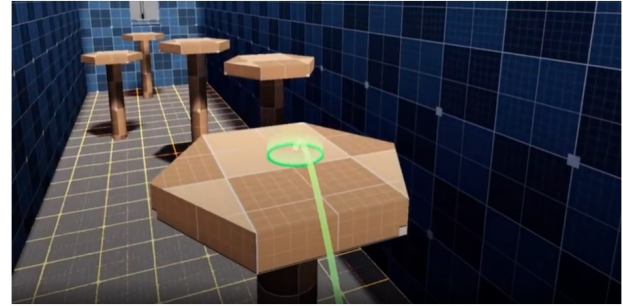


Figure 3: Visual effect highlighting the teleportation target.

training scenarios. This system ensures a structured and interactive learning experience.

4.6 Action Detection System

A state-based action detection system was designed to track user activities and evaluate their progress toward training objectives. This system records user actions, verifies task completion, and identifies errors, providing immediate feedback to enhance learning. The error detection module is particularly useful for identifying common mistakes and reinforcing correct procedures.

4.7 Sub-maps

To enable scalability and modularity, the training environments were divided into sub-maps, each representing a specific scenario or task. These sub-maps are seamlessly integrated into the broader framework, allowing for dynamic scenario switching and efficient management. The modular approach reduces development time and supports the creation of diverse training applications. Figure 4 demonstrates how sub-maps are integrated within the overall framework.

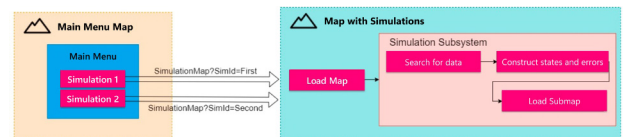


Figure 4: Sub-maps loading process.

4.8 Base project integration

The framework has been consolidated into a base project that serves as a versatile template for the rapid prototyping of VR training applications. This base project is designed with essential systems set up in a way that is both intuitive and adaptable. Developers can easily modify the meshes of interactive objects and assign various states or errors to specific actions, tailoring the functionality to their unique use cases. Additionally, they can customize the data displayed at the end of a simulation to suit specific needs.

By streamlining these processes, the framework significantly reduces development time while maintaining consistency across training scenarios. Unlike traditional development approaches, where developers often start with only the engine's basic tools and must repeatedly redesign and reimplement common systems like character controls and state tracking—this framework eliminates redundant work.

Its modular design further enhances flexibility, enabling developers to seamlessly adapt the framework to different industries and

applications. Figure 5 illustrates the default map included with the base project.

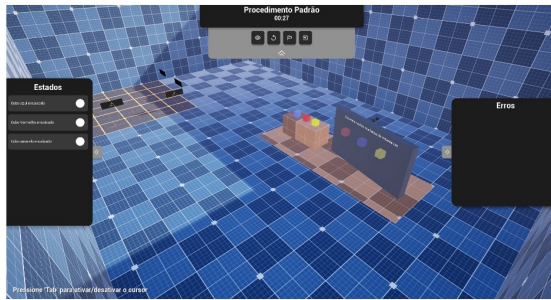


Figure 5: Default map from base project.

5 CASE STUDY

The development of the Electrical Risk Simulator was a pivotal application of the proposed framework, showcasing its potential to streamline the creation of VR training systems. The simulator was specifically designed to replicate the operational environment of an electrical substation, offering trainees a safe, immersive, and realistic platform for practicing procedures critical to their work.

Every aspect of the substation environment, from the physical layout to the intricate details of the equipment, was meticulously recreated to ensure authenticity. This high-fidelity representation not only heightened the sense of presence but also allowed users to engage deeply with the virtual environment as if they were in the real-world scenario. Figure 6 illustrates part of the electrical substation modeled for the case study.



Figure 6: Part of the modeled electrical substation.

The simulator incorporated interactive training modules tailored to various tasks within the substation, such as equipment operation, troubleshooting, and safety protocols (Figure 7). These modules utilized the framework's core features, including precise hand interactions and the snap system, enabling trainees to perform complex procedures with a high degree of accuracy (Figure 8). The error detection system played a crucial role in monitoring user actions, identifying mistakes in real time, and providing immediate feedback. This feedback mechanism was essential in reinforcing correct practices and ensuring that trainees learned effectively from their errors. Furthermore, the inclusion of dynamic locomotion options and visual indicators made navigation through the virtual environment intuitive and user-friendly.

The Electrical Risk Simulator was subjected to extensive testing to validate its functionality and effectiveness. Feedback from initial trials indicated that the system successfully met its design goals, delivering a robust training solution that could be adapted to various



Figure 7: Wearing protective equipment.



Figure 8: Handling the multimeter during the measurement procedure.

scenarios. The modularity and flexibility of the framework ensured that future updates and extensions to the simulator could be implemented seamlessly, further highlighting the framework's scalability and long-term utility.

6 FRAMEWORK VALIDATION

The simulator was evaluated to assess its impact on user performance, engagement, and overall training outcomes. A diverse group of 11 participants, ranging in age from 17 to 40 and including both professionals and students with varying levels of experience in electrical operations, took part in the evaluation under controlled conditions. Following their interaction with the simulator, participants completed the System Usability Scale (SUS) questionnaire, providing valuable insights into the system's usability and effectiveness. The simulator achieved an average SUS score of 85.9, with a standard deviation of approximately 9.64. Figure 9 captures a moment from one of the participants' test sessions.

Qualitative insights were gathered from the questionnaire results and participant interviews, emphasizing their perceptions of the simulator's usability, realism, and training relevance. The feedback was overwhelmingly positive, with participants commending the system's immersive design and user-friendly interface. Many highlighted the virtual environment's strong resemblance to real-world conditions, which boosted their confidence in transferring skills to real-life scenarios. Furthermore, interactive features—such as the snap system and the outlined indicators on interactable objects—were praised as essential elements that significantly enhanced engagement and knowledge retention.

The evaluation also revealed areas for improvement, such as incorporating additional scenarios and expanding the range of tasks to accommodate more complex training needs. These insights were instrumental in refining the framework and planning future itera-



Figure 9: User testing the Electrical Risk Simulator.

tions of the simulator. Overall, the assessment confirmed that the simulator not only met its objectives but also provided a compelling case for the broader adoption of VR-based training solutions across industries.

Through the successful deployment and evaluation of the Electrical Risk Simulator, the proposed framework demonstrated its capability to address the challenges of VR training development effectively. It reinforced the notion that well-designed VR training tools could revolutionize professional training by making it safer, more efficient, and highly engaging.

7 CONCLUSION

The development and application of the proposed framework for VR training simulators represent a significant advancement in the field of immersive learning and XR prototyping. By addressing the core challenges associated with VR training development, this work provides a robust, scalable, and user-centered solution that bridges critical gaps in the design and implementation of training environments.

One of the framework's key strengths lies in its modularity, which ensures adaptability across diverse training scenarios and industries. By compartmentalizing functionalities into reusable systems, such as interaction handling, error detection, and locomotion, the framework not only reduces development time but also enables customization to meet specific training objectives. This modularity was exemplified in the Electrical Risk Simulator, where complex tasks were seamlessly integrated, delivering a highly realistic and intuitive experience for trainees.

Another standout feature of the framework is its emphasis on user experience. The incorporation of intuitive interaction systems, dynamic feedback mechanisms, and immersive locomotion techniques ensures that the training process is both engaging and effective. The ability to detect errors in real time and provide actionable feedback empowers users to learn actively, reinforcing correct behaviors while mitigating the risks of repetitive mistakes. The focus on user-centric design principles enhances the framework's accessibility, making it suitable for novice and non-technical users, a priority highlighted in the workshop's objectives.

The application of the framework in the Electrical Risk Simulator also demonstrated its capacity to replicate high-fidelity environments for critical training scenarios. The detailed recreation of a substation environment, combined with interactive modules and realistic task simulations, resulted in a platform that is not only effective for skill acquisition but also scalable for future expansions. Quantitative evaluations, such as reductions in task completion times and error rates, coupled with positive qualitative feedback from participants, underline the framework's effectiveness in improving training outcomes.

This work also contributes to the broader discourse on XR prototyping by demonstrating how frameworks can unify common development challenges into a cohesive structure. The emphasis on modularity, scalability, and user experience provides a template for future research and development in the field. Moreover, the framework's adaptability to various XR technologies, including potential extensions to augmented reality (AR) and mixed reality (MR), positions it as a versatile tool for advancing the state-of-the-art in immersive training solutions.

Despite these achievements, the work also highlights opportunities for future exploration. Expanding the framework to accommodate a broader range of scenarios and integrating generative AI tools for content creation could further enhance its capabilities. Additionally, conducting longitudinal studies across different industries would provide deeper insights into its long-term impact on learning efficiency and professional training practices.

In conclusion, this work not only provides a practical solution to the challenges of VR training development but also establishes a foundation for future advancements in XR prototyping techniques. The strong results achieved through the application of the framework in real-world scenarios affirm its potential to revolutionize immersive training, bridging the gap between cutting-edge technology and effective skill development. As XR technologies continue to evolve, this framework offers a pathway for creating impactful, scalable, and transformative solutions in the realm of immersive learning.

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